

Report for Mandatory Assignment 1

Tobias Meyer Andersen

October 2026

1 Problem 1.2.3

Problem: show that $e^{i(k_x x + k_y y - \omega t)}$ satisfies $u_{tt} = c^2(u_{xx} + u_{yy})$.

1.1 Solution

Let us define some variables that will help us express the algebra more concisely:

$$\mathcal{X} = \cos(k_x x) + i \sin(k_x x) \quad (1)$$

$$\mathcal{Y} = \cos(k_y y) + i \sin(k_y y) \quad (2)$$

$$\mathcal{T} = \cos(-\omega t) + i \sin(-\omega t) \quad (3)$$

$$e^{i(k_x x + k_y y - \omega t)} = \mathcal{X}\mathcal{Y}\mathcal{T} \quad (4)$$

We can compute the derivatives in polar form and use the symbols we just defined to make the expression more manageable. The double derivatives in space and times are shown in the subsequent three equations.

$$u_{xx} = (-k_x^2 \cos(k_x x) - k_x^2 i \sin(k_x x))\mathcal{Y}\mathcal{T} = -k_x^2 \mathcal{X}\mathcal{Y}\mathcal{T} \quad (5)$$

$$u_{yy} = (-k_y^2 \cos(k_y y) - k_y^2 i \sin(k_y y))\mathcal{X}\mathcal{T} = -k_y^2 \mathcal{X}\mathcal{Y}\mathcal{T} \quad (6)$$

$$u_{tt} = (-\omega^2 \cos(\omega t) - \omega^2 i \sin(\omega t))\mathcal{X}\mathcal{Y} = -\omega^2 \mathcal{X}\mathcal{Y}\mathcal{T} \quad (7)$$

Inserting this into the wave equation we have:

$$-\omega^2 \mathcal{X}\mathcal{Y}\mathcal{T} = c^2(-k_x^2 \mathcal{X}\mathcal{Y}\mathcal{T} - k_y^2 \mathcal{X}\mathcal{Y}\mathcal{T}) \quad (8)$$

Simplifying this leads directly to the following:

$$\omega^2 = c^2(k_x^2 + k_y^2) \quad (9)$$

The proof is concluded by having the wave equation be satisfied precisely when the dispersion ω satisfies:

$$\omega = c\sqrt{k_x^2 + k_y^2} \quad (10)$$

2 Problem 1.2.4

Problem: prove that the numeric dispersion is equal to the exact one for CFL value of $1/\sqrt{2}$.

We have the finite difference-discretized 2 wave equation from Equation 1.3 in the problem set:

$$\frac{U_{ij}^{n+1} - 2U_{ij}^n + U_{ij}^{n-1}}{\Delta t^2} = \frac{c^2}{h^2} (U_{i+1,j}^n + U_{i-1,j}^n + U_{i,j+1}^n + U_{i,j-1}^n - 4U_{ij}^n). \quad (11)$$

which we simplify using the $k_x = k_y$ from the problem statement:

$$\frac{U_{ij}^{n+1} - 2U_{ij}^n + U_{ij}^{n-1}}{\Delta t^2} = \frac{c^2}{h^2} (2U_{i+j+1}^n + 2U_{i+j-1}^n - 4U_{ij}^n). \quad (12)$$

The courant number is defined as:

$$C = \frac{c \Delta t}{h}.$$

With the following assumption from the problem statement:

$$C = \frac{1}{\sqrt{2}}, \quad C^2 = \frac{c^2 \Delta t^2}{h^2} = \frac{1}{2}. \quad (13)$$

Rearranging Equation 12 we get:

$$\frac{U_{ij}^{n+1} - 2U_{ij}^n + U_{ij}^{n-1}}{2U_{i+j+1}^n + 2U_{i+j-1}^n - 4U_{ij}^n} = \frac{c^2 \Delta t^2}{h^2} = \frac{1}{2} \quad (14)$$

This can now be simplified to:

$$U_{ij}^{n+1} - 2U_{ij}^n + U_{ij}^{n-1} = U_{i+j+1}^n + U_{i+j-1}^n - 2U_{ij}^n \quad (15)$$

Subtracting $2U_{i+j}^n$ from both sides leaves us with:

$$U_{ij}^{n+1} + U_{ij}^{n-1} = U_{i+j+1}^n + U_{i+j-1}^n \quad (16)$$

Making further simplifications and progressing the proof requires using the exponential form of these points instead, from the problem statement we have

$$U_{ij}^n = e^{i(kh(i+j) - \tilde{\omega}n\Delta t)}, \quad (17)$$

where $\tilde{\omega}$ is the numerical angular frequency.

Inserting this into Equation 16 produces:

$$e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} + e^{i(kh(i+j) - \tilde{\omega}(n-1)\Delta t)} = e^{i(kh(i+j+1) - \tilde{\omega}n\Delta t)} + e^{i(kh(i+j-1) - \tilde{\omega}n\Delta t)} \quad (18)$$

To start cancelling out the exponentials we multiply first by $\frac{1}{e^{i(kh(i+j) - \tilde{\omega}n\Delta t)}}$, and then $\frac{1}{e^{-i\tilde{\omega}n\Delta t}}$:

$$e^{-i\tilde{\omega}\Delta t} + e^{i\tilde{\omega}n\Delta t} = e^{ikh} + e^{-ikh} \quad (19)$$

Further simplifications are available if we now express this in polar form instead:

$$\cos(kh) + i\sin(kh) + \cos(-kh) + i\sin(-kh) = \cos(\tilde{\omega}\Delta t) + i\sin(\tilde{\omega}\Delta t) + \cos(-\tilde{\omega}\Delta t) + i\sin(-\tilde{\omega}\Delta t) \quad (20)$$

Using the trigonometric identities $\cos(-x) = \cos(x)$ and $\sin(-x) = -\sin(x)$ we can cancel the sines and add the cosines, which results in the following equation:

$$2\cos(kh) = 2\cos(\tilde{\omega}\Delta t) \quad (21)$$

Solving for $\tilde{\omega}$ we obtain:

$$\tilde{\omega} = \frac{kh}{\Delta t} \quad (22)$$

Now that we have an expression for the numerical dispersion we can retrieve from an earlier problem exact dispersion:

$$\omega = c\sqrt{k_x^2 + k_y^2} \quad (23)$$

Using the fact that in our problem $k_x = k_y$ and squaring both sides we are left with this analytic statement:

$$\omega^2 = 2c^2k^2 \quad (24)$$

We now have an expression for both the numeric and the exact dispersion, and the goal is to prove that they are equal. We create the following statement:

$$\omega^2 = 2c^2k^2 = \frac{k^2h^2}{\Delta t^2} = \tilde{\omega}^2 \quad (25)$$

Cancelling k^2 from both sides we have that the lhs and rhs are equal whenever $2c^2 = \frac{h^2}{\Delta t^2}$. Putting all variables on the same side we have:

$$\frac{1}{2} = \frac{c^2\Delta t^2}{h^2} \quad (26)$$

Taking the square root on both sides we have:

$$\frac{1}{\sqrt{2}} = \frac{c\Delta t}{h} \quad (27)$$

This concludes the proof, as we have now shown that the numerical dispersion is exact whenever we have a CFL value of $\frac{1}{\sqrt{2}}$, which was an initial assumption.